

Total No. of Questions : 8]

SEAT No. :

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[6583]-43

T.E. (Computer/Computer Science & Engineering(Data Science)/

Computer Science)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - V) (310242)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Give Context Free Grammars for the following languages and show the derivation for given string [12]

- i) $L = \{ w \in \{a,b\}^* \mid w \text{ is string of starting with 'a' and ending with 'b'} \}$
show the derivation for "ababab"
- ii) $L = a^n b^{2n}$ where $n \geq 1$. Show the derivation for "aabbabb"
- iii) $RE = (0+1)^*$ Show the derivation for "0110"

b) Simplify the following grammar as [5]

- i) Eliminate Useless production

$S \rightarrow aAa|bBb| \epsilon$

$A \rightarrow C|a$

$B \rightarrow C|b$

$C \rightarrow CDE| \epsilon$

$D \rightarrow A|B|ab$

- ii) Eliminate Unit Production

$S \rightarrow 0A | 1B | C$

$A \rightarrow 0S | 00$

$B \rightarrow 1 | A$

$C \rightarrow 01$

OR

P.T.O.

Q2) a) The regular grammar is also called a right-linear grammar. But Every language generated from a left-linear grammar is regular. Why? Prove that Every language generated from a left-linear grammar is regular with the following grammar examples. [8]

i) $G = (\{S, A\}, \{0, 1\}, \{S \rightarrow A01, A \rightarrow 10\}, S)$

ii) $G = (\{S, A\}, \{0, 1\}, \{S \rightarrow A1, A \rightarrow A0|\epsilon\}, S)$

b) i) Show the derivation tree for string 'aabbabb' with the following grammar. [3]

$S \rightarrow AB | \epsilon$

$A \rightarrow aB$

$B \rightarrow Sb$

ii) Construct a derivation tree for the string aabbabba for the CFG given by, [3]

$S \rightarrow aB | bA$

$A \rightarrow a | aS | bAA$

$B \rightarrow b | bS | aBB$

iii) Check whether the given grammar G is ambiguous or not. [3]

$A \rightarrow AA$

$A \rightarrow (A)$

$A \rightarrow a$

Q3) a) i) Construct PDA for the given CFG and test whether 010^4 is acceptable by this PDA.

$S \rightarrow 0BB$

$B \rightarrow 0S | 1S | 0$

ii) Construct PDA for the given CFG and test whether 'aaabb' is acceptable by this PDA.

$S \rightarrow aSb$

$S \rightarrow a | b | \epsilon$

[10]

b) Draw a PDA for the CFG given below :

$S \rightarrow aSb$

$S \rightarrow a | b | \epsilon$

Show the simulation of string acceptable by above CFG using PDA transitions [8]

OR

- Q4) a)** Give a PDA for the following language [10]
 $L = \{ \text{the set of strings over alphabet } \{a,b\} \text{ with exactly twice as many } a\text{'s as } b\text{'s} \}$ give the demonstration for the string “abaabbaaaaab”
- b)** Design a PDA for detection of palindrome over $\{a, b\}$ for even and odd length both. [8]

- Q5) a)** Construct a TM for the language $L = \{0^n 1^n 2^n\}$ where $n \geq 1$ Show simulation for “001122” [10]
- b)** Write short notes on : [8]
- Halting Problem of Turing Machine
 - Representation of Turing Machine
 - Multi-tape Turing Machine
 - Instantaneous Description of a Turing Machine

OR

- Q6) a)** Construct TM for the addition function for the unary number system. And show the simulation [9]
 For example
 $2 + 3$
 i.e. $11 + 111 = 11111$
- b)** Draw FA and Corresponding TM With for language L of the following description $L = \{x \in \{a, b\}^* \mid x \text{ ends with } aba\}$ [9]

- Q7) a)** What Minimum spanning tree problem? Prove that finding MST by using Kruskal’s algorithm is in class P. [5]
- b)** What is post correspondence problem? Why is post correspondence problem undecidable? Explain PCP with following instance of the set of the strings A and B [12]

	A	B
1.	1	111
2.	10111	10
3.	10	0

OR

Q8) a) Define the Class P and Class NP and Problem with their example in detail. **[8]**

b) Find whether the following lists have a Post Correspondence, if yes give the sequence **[9]**

i) $M = (abb, aa, aaa)$ and $N = (bba, aaa, aa)$

ii) $M = (B, A, CA, ABC)$ and $N = (CA, AB, A, C)$
